

Question	Answer	Marks
3(a)	mass is the property (of a body/object) resisting changes in motion or mass is the quantity of matter (in a body)	B1
3(b)(i)	force on A (by B) equal <u>and opposite</u> to force on B (by A) or both A and B exert equal and opposite forces on each other	B1
	force is rate of change of momentum <u>and</u> time (of contact) is same	B1
3(b)(ii)	$p = mv$ or $3M \times 0.40$ or $M \times 0.25$ or $3M \times 0.2$ or Mv	C1
	$(3M \times 0.40) - (M \times 0.25) = (3M \times 0.2) + Mv$	C1
	$v = (3 \times 0.40) - 0.25 - (3 \times 0.2)$ $= 0.35\text{ms}^{-1}$	A1
3(b)(iii)	1. relative speed of approach = $0.40 + 0.25$ $= 0.65\text{ms}^{-1}$	A1
	2. relative speed of separation = $0.35 - 0.20$ $= 0.15\text{ms}^{-1}$	A1
3(b)(iv)	(relative) speed of separation not equal to/less than (relative) speed of approach or answers (to (b)(iii) are) not equal and so inelastic collision	B1

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1(c)(i)	time for one oscillation/one vibration/one cycle	B1